Reminders and announcements

- During lectures you can submit questions to: vevox.app ID: 107-018-952
- You can find answers to questions in the COMP1870 Teams Channel (link in Minerva) (along with other discussion).
- Maths Lunch Hours, Mondays and Wednesdays, 1.00pm - 2.00pm. Bragg GR.18.
- My student hour: Tuesdays 1.00pm - 2.00pm, Bragg building foyer.

Reminders and announcements

Starting on Thursday 4 November, COMP1870 lectures change to:

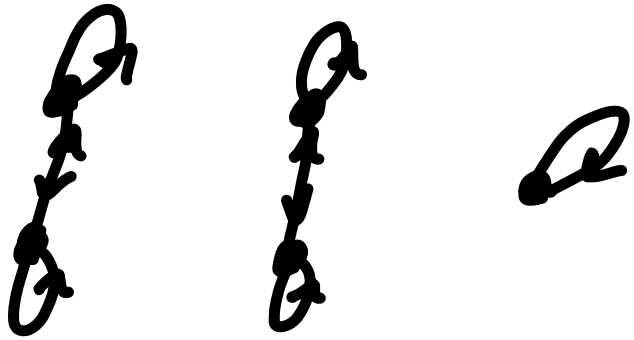
- Thursdays from 1.00pm to 3.00pm in
  Chem. & Process Engineering Lecture Theatre A (2.23)
  (Starting from Thursday 4 November)
- Mondays from 4.00pm to 6.00pm in
  Mechanical Engineering Lecture Theatre A (1.33)
  (Starting from Monday 9 November)

Revision

- A **relation** from A to B is any subset $R \subseteq A \times B$.
- We write aRb to mean that (a, b) is in R.
- If $A = B$, then R is a **relation on** A. $R \subseteq A \times A$

Property	**Definition**	**Characterisation via arrows**
Reflexive $\forall a$	$(a, a) \in R$ for all $a \in A$	Every element has a loop
Symmetric $\forall a,b$	$(a, b) \in R$ implies $(b, a) \in R$	Every arrow is two-way
Anti-symmetric $\forall a,b$	$(a, b) \in R$ and $(b, a) \in R$ implies $a = b$	The only two-way arrows are loops
Transitive $\forall a,b,c$	$(a, b) \in R$ and $(b, c) \in R$ implies $(a, c) \in R$	If we can get from a to c in two steps, we can get there in one step

Revision



Let A be a set.

- An **equivalence relation** on A is a relation on A that is reflexive, symmetric, and transitive.
- If R is an equivalence relation on A and a is an element of A, then $[a] = \{x \in A : aRx\}$ is the **equivalence class** containing a.
- The equivalence classes **partition** A. (This means that each element of A is in exactly one equivalence class.)

Partial orders

Definition 7.11. Let A be a set and let $R \subseteq A \times A$ be a relation on A. If R is reflexive, anti-symmetric, and transitive, then R is a **partial order**.

$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \leq y\}$ is a partial order.

Proposition 7.12. Let A be a set and let $R \subseteq \mathcal{P}(A) \times \mathcal{P}(A)$ be the relation on subsets of A such that (X, Y) is in R if and only if X is a subset of Y. Then R is a partial order.

Hasse diagrams

Partial orders can be represented by diagrams called **Hasse diagrams**.

A Hasse diagram can be obtained from the directed graph diagram by the following steps.

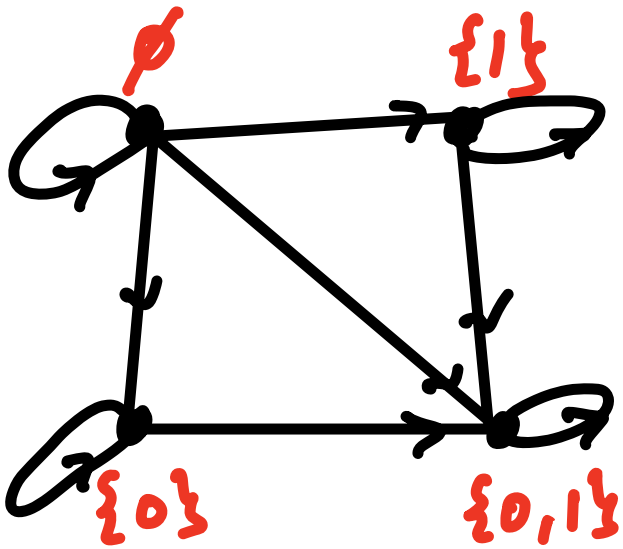
- Since the relation is reflexive, we know that every element has a loop. Therefore we can omit loops.
- Since the relation is anti-symmetric it cannot contain any **directed cycles** (Proposition 7.13). So we can draw the directed graph with every arrow directed upwards on the page. Therefore we can omit arrowheads.
- Since the relation is transitive, we can omit any arrows implied by transitivity.



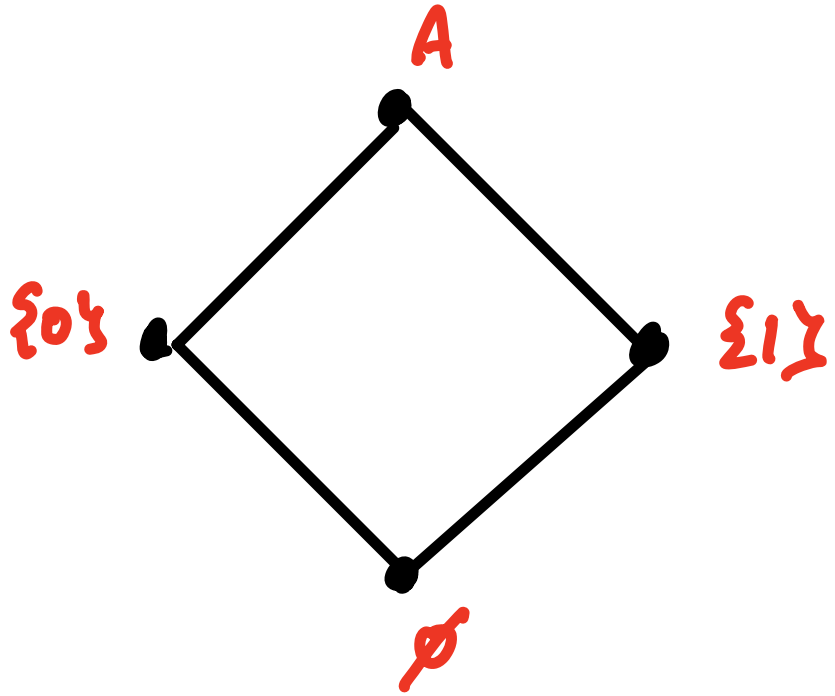
Hasse diagrams

Let $A = \{0, 1\}$.

Let $R = \{(X, Y) \in \mathcal{P}(A) \times \mathcal{P}(A) : X \subseteq Y\}$



Directed graph



Hasse diagram

Cardinality

The number of elements in a set is called the **cardinality** or the **size** of the set.

If A is a set, then the cardinality of A is denoted by $|A|$. (Alternative notation: $\#A$.)

If $A = \{a, b, c, d\}$, then $|A| = 4$

If $A = \{0\}$, then $|A| = 1$

If $A = \{\emptyset\}$, then $|A| = 1$

If $A = \emptyset$, then $|A| = 0$

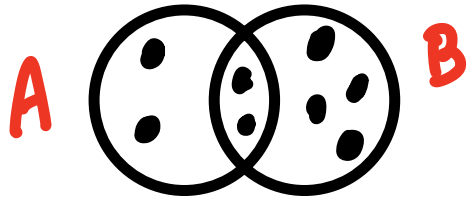
Addition rule

Let A and B be sets. If $A \cap B$ is empty, then we say that A and B are **disjoint**.

Proposition 8.3. Let A and B be disjoint finite sets. Then

$$|A \cup B| = |A| + |B|$$

This is known as the **addition rule** (or addition principle). Note that it does not hold if $A \cap B$ is non-empty.



$$|A \cup B| = 8 \neq 4 + 6 = |A| + |B|$$

Say that $A_1, A_2, \ldots, A_m$ are sets. If each pair of these sets is disjoint, then we say they are **pairwise disjoint**. In this case,

$$|A_1 \cup A_2 \cup \cdots \cup A_m| = |A_1| + |A_2| + \cdots + |A_m|$$

Multiplication rule

Let A and B be sets. Then $A \times B = \{(a, b) \mid a \in A, b \in B\}$ is the **Cartesian product** of A and B.

Proposition 8.4. Let A and B be finite sets. Then

$$|A \times B| = |A| \cdot |B|$$

This is known as the **multiplication rule** (or principle).

Say that $A_1, A_2, \ldots, A_m$ are sets. Then $A_1 \times A_2 \times \cdots \times A_m$ is the set of all tuples $(a_1, a_2, \ldots, a_m)$ such that a_i is an element of A_i for each possible i. Then

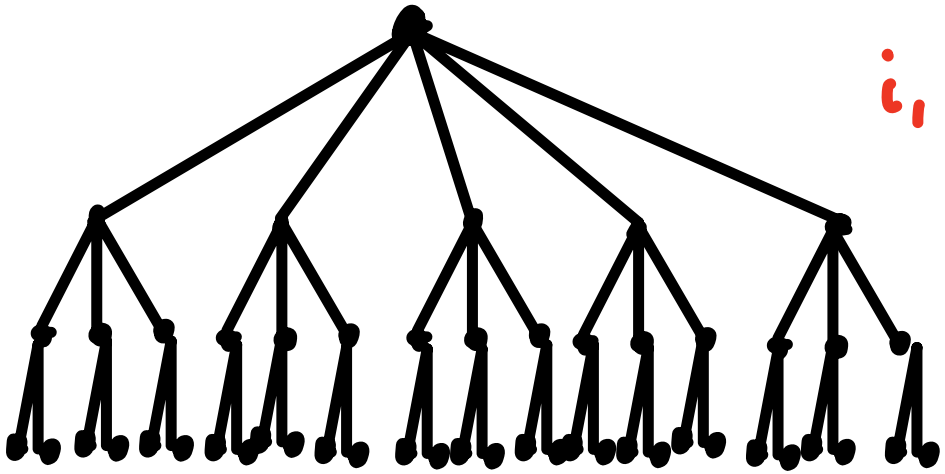
$$|A_1 \times A_2 \times \cdots \times A_m| = |A_1| \cdot |A_2| \cdot \ldots \cdot |A_m|$$

Multiplication rule

The multiplication rule says:

> *Suppose we have n choices to make. Assume that for the first choice we can choose from i_1 options, for the second choice we can choose from i_2 options, and so on, so that for choice number j, we have i_j options. The total number of ways we can make the choices is*

$$i_1 \times i_2 \times \cdots \times i_n$$



$i_1 = 5$

$i_2 = 3$

$i_3 = 2$

Number of paths from root to a leaf $5 \times 3 \times 2 = 30$

Applications of the multiplication rule

Remember that when A is a set, $\mathcal{P}(A)$ is the **power set** of A: the set of all subsets of A.

Proposition 8.5. Let A be a finite set such that $|A| = n$. Then $|\mathcal{P}(A)| = 2^n$

Proof. We make a sequence of n choices: one for each element of A. We are choosing a subset $X \subseteq A$, and we consider each element of A in turn. For each choice, we have two options: the element is in X, or it is not. The multiplication principle says we have $\underbrace{2 \times 2 \times \cdots \times 2}_{n \text{ times}}$ possible ways of choosing a subset. $\square$

Permutations

Let A be a set. A **permutation** of A is an ordering of the elements of A. When n is a ~~non-negative~~ positive integer, **n factorial** is

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1.$$

It is convenient to define 0! to be 1.

Proposition 8.6. Let A be a set with $|A| = n$. The number of permutations of A is $n!$.

Proof. The number of options for the 1st element is n. The number of options for 2nd is n-1, etc. □

Ordered selections

Let A be a set. When we make an **ordered selection** from A, we choose elements from A, and the order in which we choose them is **significant**. We make the selection without repetition.

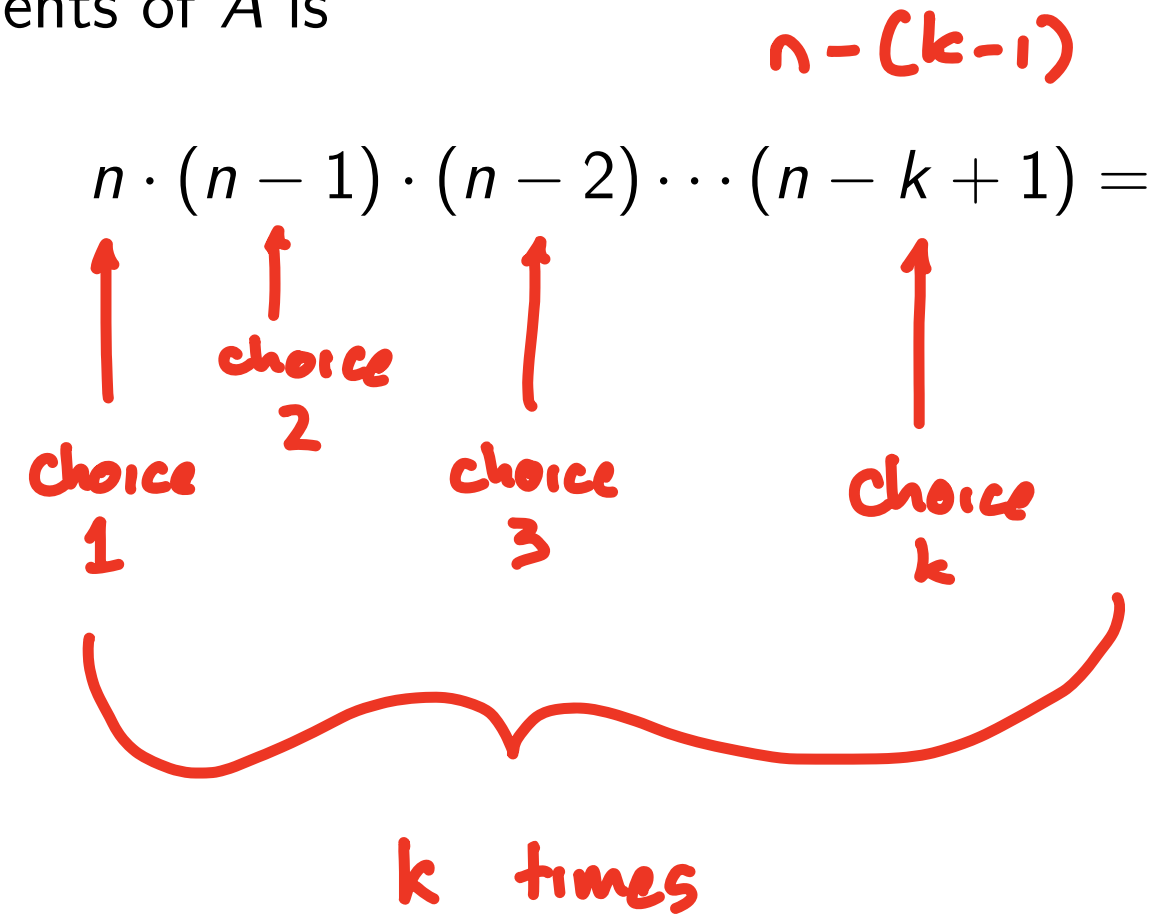
Example. Say that $A = \{1, 2, 3, 4, 5\}$. In an ordered selection, choosing 3 first and 4 second is different from choosing 4 first and 3 second.

If $A = \{a, b, c, d\}$, the ordered selections of 2 elements are: ab, ac, ad, bc, bd, cd, ba, ca, da, cb, db, dc

Ordered selections

Proposition 8.7. Let A be a set with $|A| = n$. The number of ways we can make an ordered selection of k distinct elements of A is

$$n \cdot (n-1) \cdot (n-2) \cdots (n-k+1) = \frac{n!}{(n-k)!}.$$



Unordered selections

Let A be a set. When we make an **unordered selection** from A, we choose elements from A, and the order in which we choose them is **not significant**. We do not allow repetition, so the elements are distinct.

Example. Say that $A = \{1, 2, 3, 4, 5\}$. In an ordered selection, choosing 3 first and 4 second is the same as choosing 4 first and 3 second.

Example. We want to choose a committee of 3 people from a class of 10. There are:

$$\frac{10!}{(10-3)!\ 3!} = \frac{10 \times 9 \times 8 \times 7 \times \cdots \times 2 \times 1}{(7 \times 6 \times \cdots \times 2 \times 1)(3 \times 2 \times 1)} = 120$$

Binomial coefficients

Proposition 8.8. Let A be a set with $|A| = n$. The number of ways we can make an unordered selection of k distinct elements of A is

$$\frac{n!}{(n-k)!k!}.$$

Sketch proof: There are $\frac{n!}{(n-k)!}$ ways of making an ordered selection. Each ordered selection is represented $k!$ times, since this is the number of ways of ordering k elements. So we divide $\frac{n!}{(n-k)!}$ by $k!$ to compensate. □

Binomial coefficients

Proposition 8.8. Let A be a set with $|A| = n$. The number of ways we can make an unordered selection of k distinct elements of A is

$$\frac{n!}{(n-k)!k!}.$$

This is equal to the number of subsets of A with size k.

The number $\dfrac{n!}{(n-k)!k!}$ is denoted by $\binom{n}{k}$ and is called "n **choose** k". These numbers are called **binomial coefficients**.

Alternative notation: ${}^{n}C_{k}$

The binomial theorem

$$
\begin{aligned}
(a+b)^0 &= 1 \\
(a+b)^1 &= 1a + 1b \\
(a+b)^2 &= 1a^2 + 2ab + 1b^2 \\
(a+b)^3 &= 1a^3 + 3a^2b + 3ab^2 + 1b^3 \\
(a+b)^4 &= 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4 \\
(a+b)^5 &= 1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5
\end{aligned}
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(a+b)^5 &= 1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5
\end{aligned}
$$

$$
\begin{aligned}
(a+b)^0 &= \binom{0}{0} \\
(a+b)^1 &= \binom{1}{0}a + \binom{1}{1}b \\
(a+b)^2 &= \binom{2}{0}a^2 + \binom{2}{1}ab + \binom{2}{2}b^2 \\
(a+b)^3 &= \binom{3}{0}a^3 + \binom{3}{1}a^2b + \binom{3}{2}ab^2 + \binom{3}{3}b^3 \\
(a+b)^4 &= \binom{4}{0}a^4 + \binom{4}{1}a^3b + \binom{4}{2}a^2b^2 + \binom{4}{3}ab^3 + \binom{4}{4}b^4 \\
(a+b)^5 &= \binom{5}{0}a^5 + \binom{5}{1}a^4b + \binom{5}{2}a^3b^2 + \binom{5}{3}a^2b^3 + \binom{5}{4}ab^4 + \binom{5}{5}b^5
\end{aligned}
$$

The binomial theorem

Theorem 8.9. Let n be a non-negative integer. Then

$$(a+b)^n = \binom{n}{0} a^n b^0 + \binom{n}{1} a^{n-1} b^1 + \cdots + \binom{n}{n-1} a^1 b^{n-1} + \binom{n}{n} a^0 b^n$$

$$= \sum_{i=0}^{n} \binom{n}{i} a^{n-i} b^i$$

$(a+b)^4 = (a+b)(a+b)(a+b)(a+b) = abab + baab$

Pascal's Triangle

This array is called **Pascal's Triangle**.

$$\begin{array}{ccccccccccc}
 & & & & \binom{0}{0} & & & & & & \\
 & & & \binom{1}{0} & & \binom{1}{1} & & & & & \\
 & & \binom{2}{0} & & \binom{2}{1} & & \binom{2}{2} & & & & \\
 & \binom{3}{0} & & \binom{3}{1} & & \binom{3}{2} & & \binom{3}{3} & & & \\
\binom{4}{0} & & \binom{4}{1} & & \binom{4}{2} & & \binom{4}{3} & & \binom{4}{4} & & \\
 & \binom{5}{1} & & \binom{5}{2} & & \binom{5}{3} & & \binom{5}{4} & & \binom{5}{5} &
\end{array}$$

$$\begin{array}{ccccccccccc}
 & & & & 1 & & & & & & \\
 & & & 1 & & 1 & & & & & \\
 & & 1 & & 2 & & 1 & & & & \\
 & 1 & & 3 & & 3 & & 1 & & & \\
1 & & 4 & & 6 & & 4 & & 1 & & \\
 & 5 & & 10 & & 10 & & 5 & & 1 &
\end{array}$$

1 6 15 20 15 6 1

Pascal's Identity

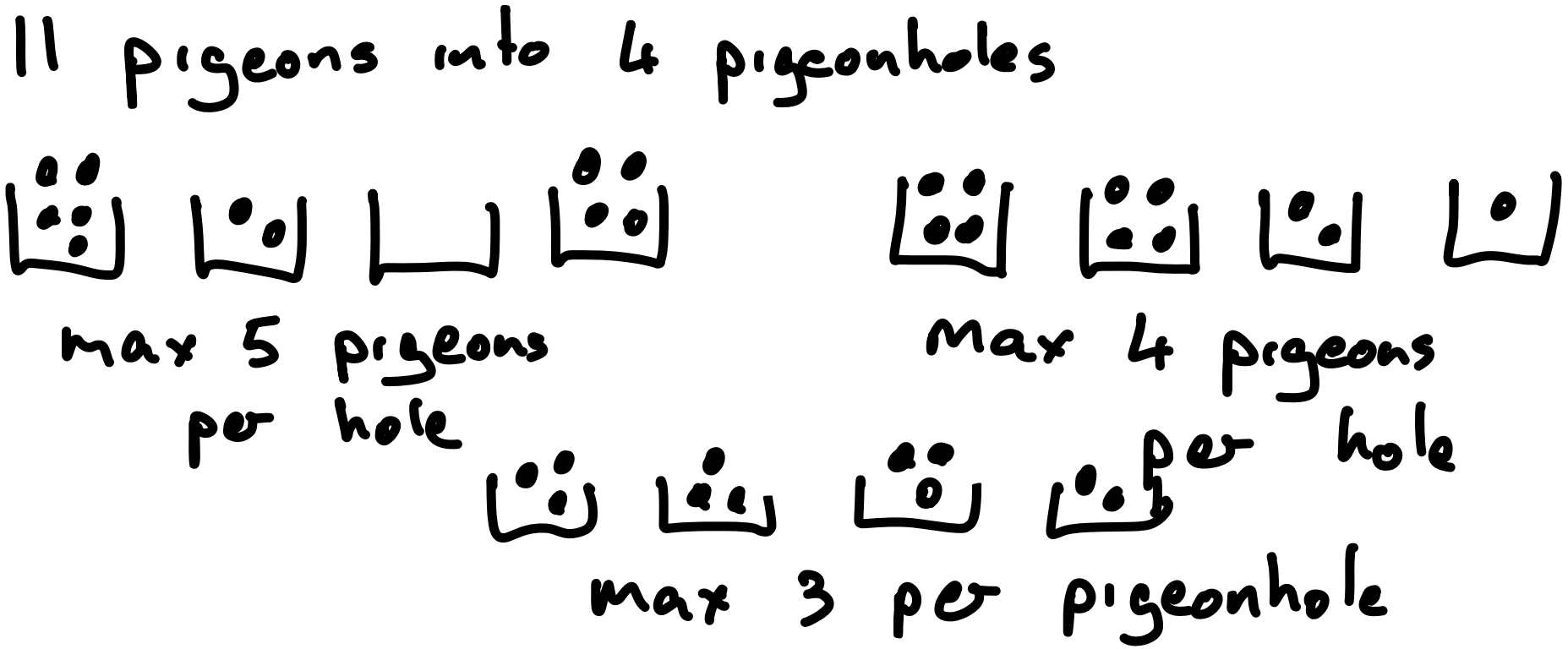
Proposition 8.10. Let n and k be positive integers where $k < n$. Then

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

The pigeonhole principle

Imagine that we distribute n pigeons into k pigeonholes. There may be many such distributions.

We examine these distributions and look at the maximum number of pigeons in a pigeonhole.



Can we give a lower bound on this maximum?